Tharun Kumar Poongan

ECE Undergraduate | Embedded Systems | Automotive Electronics & Hardware
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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate (2027 batch) with practical exposure to embedded firmware development, microcontroller architectures, sensor interfacing, and automotive monitoring systems. Proficient in C, Embedded C, and circuit analysis, with hands-on project experience in EV thermal safety cutoff logic, multi-channel ADC sampling, I2C telemetry, and hardware-software validation. Seeking Graduate Engineer Trainee and Core Electronics roles across automotive, manufacturing, and embedded sectors.

EDUCATION

B.E. in Electronics and Communication Engineering **Expected 2027**
Government College of Engineering, Erode — Anna University CGPA: 7.79

TECHNICAL SKILLS

Embedded Systems & MCUs: Embedded C, ESP32, Arduino, Microcontroller Firmware, ADC Sampling, Safe-State Logic

Protocols & Interfacing: I2C, Sensor Interfacing, Multi-channel Telemetry, Network Sockets, Multi-threading

Hardware & Electronics: Circuit Design, Circuit Analysis, Schematic Evaluation, PCB Design Basics, KiCad (Basic)

Programming Languages: C, Embedded C, C++ (Learning)

Testing & Tools: Wokwi Simulation, Hardware Debugging, Testbenches, Linux (Basic), Git, GitHub, Cursor

INTERNSHIP EXPERIENCE

Embedded C & Microcontroller Intern **June 2026**
Ingage Technologies Private Limited — Virtual Internship

- Developed modular Embedded C firmware routines targeting microcontroller hardware and peripheral sensor interfaces.
- Implemented multi-channel ADC sampling routines for parameter monitoring and continuous threshold limit verification.
- Programmed deterministic safe-state shutdown routines and validated system execution stability in simulation testbenches.

Full Stack AI Intern **Jun – Jul 2025**
Nuvai AI Solutions Pvt Ltd (Two47 Letters Tech), Bengaluru, India

- Supported backend system validation, automated data routing, and technical verification workflows.
- Prepared technical testing documentation, validation procedures, and module-level defect traceability records.
- Utilized Git and GitHub for collaborative branch management, code verification, and version-controlled updates.

CORE TECHNICAL PROJECTS

Smart EV Battery Pack Safety & Thermal Monitoring System **Individual Project**
Embedded C, ESP32, Wokwi, Temperature Sensor, I2C

- Engineered an automotive EV battery safety and thermal monitoring system using Embedded C on an ESP32 microcontroller.
- Interfaced temperature sensors and analog inputs via I2C and ADC to execute automated safe-state cutoff logic during thermal runaways.
- Evaluated circuit schematics and verified continuous sensor communication routines inside the Wokwi simulation platform.

Real-Time V2V Communication Simulator **Hackathon Team Project**
C/C++, Network Sockets, Multi-threading

- Built a multi-node telemetry and vehicle-to-vehicle (V2V) communication simulation utilizing C/C++ and network sockets.
- Analyzed real-time packet propagation, payload serialization, and queue constraints to assess communication reliability.
- Implemented multi-threaded processing to sustain concurrent node data exchange without packet loss under high load.

KEY ACHIEVEMENTS

- **2nd Place, National Hackathon** — Presidency University, Bangalore (2024); Cash Award: INR 15,000
- **Innovation Award, HackTronix 1.0** — Sri Sairam Engineering College (2025); Cash Award: INR 2,000